

BOGORODSKOYE, Russian Federation

WMO: 314390

Lat: 52.3731N

Lon: 140.4330E

Elev: 114

StdP: 14.64

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.8	-26.3	-34.9	0.8	-29.5	-31.2	1.0	-26.1	27.8	5.0	23.4	3.6	1.4	160	0.600

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.8	80.7	68.7	77.8	66.9	75.3	65.3	71.1	78.1	68.8	75.5	66.9	73.3	5.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
68.5	105.0	75.9	66.2	97.0	73.1	64.2	90.3	71.0	35.1	77.9	33.2	75.7	31.6	73.5	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
19.7	16.6	13.9	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-34.8	86.4	4.9	3.3	-38.4	88.8	-41.3	90.7	-44.1	92.6	-47.7	95.0		
			-34.0	74.5	5.6	3.0	-38.0	76.6	-41.3	78.3	-44.5	80.0	-48.5	82.2		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	31.0	-8.4	-2.8	13.3	31.9	47.0	58.4	65.2	64.1	53.6	37.6	14.2	-4.6
	DBStd	27.82	11.35	10.06	9.43	7.61	7.09	6.54	5.40	4.81	6.48	7.12	12.21	11.97
	HDD50	8316	1810	1478	1138	543	145	9	0	0	37	388	1075	1693
	HDD65	12558	2275	1898	1603	992	559	213	65	77	343	850	1525	2158
	CDD50	1368	0	0	0	0	51	261	471	436	146	3	0	0
	CDD65	138	0	0	0	0	0	16	72	48	2	0	0	0
	CDH74	848	0	0	0	0	13	157	441	230	7	0	0	0
	CDH80	110	0	0	0	0	1	18	72	19	0	0	0	0
Wind	WSAvg	5.1	5.3	5.5	5.8	5.6	5.7	4.7	4.4	4.3	4.3	5.1	5.1	5.0
Precipitation	PrecAvg	20.5	1.1	0.6	0.7	1.0	1.9	2.0	2.2	3.8	2.4	2.0	1.4	1.4
	PrecMax	27.8	2.5	2.1	1.7	1.9	3.5	6.3	4.7	7.5	5.2	5.0	3.2	3.6
	PrecMin	10.9	0.0	0.0	0.1	0.2	0.4	0.4	0.0	0.2	0.1	0.4	0.3	0.1
	PrecStd	3.9	0.7	0.5	0.5	0.4	0.9	1.3	1.3	1.9	1.1	1.1	0.7	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.8	27.5	39.6	59.7	75.2	82.3	85.4	82.4	74.9	60.9	41.9	31.4
		MCWB	28.7	25.5	32.8	46.7	58.0	67.2	71.7	72.2	64.8	51.6	38.5	30.1
	2%	DB	23.3	21.6	34.5	52.2	69.3	78.1	81.8	79.1	71.2	55.4	37.0	27.0
		MCWB	22.4	19.8	29.8	42.0	54.8	64.8	69.2	69.4	62.9	49.1	34.1	25.8
	5%	DB	14.9	16.8	31.4	47.2	64.9	74.9	78.6	76.6	67.6	51.4	34.0	20.9
		MCWB	14.0	15.1	27.9	38.9	51.9	62.8	67.3	67.7	60.3	45.9	31.5	20.1
	10%	DB	8.8	12.5	28.6	43.2	60.6	71.5	75.9	73.7	64.3	48.3	31.2	12.6
		MCWB	7.8	11.0	25.5	36.5	49.7	60.8	66.1	65.9	58.4	43.8	28.9	11.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.6	25.5	33.6	47.6	59.8	70.2	74.3	74.1	67.0	53.4	38.6	30.5
		MCDB	29.7	27.0	38.1	57.4	71.9	79.6	82.8	79.9	72.4	58.1	41.4	31.5
	2%	WB	22.2	19.6	30.5	42.9	56.1	66.6	71.2	71.4	63.7	49.9	34.6	25.7
		MCDB	23.0	21.3	33.9	50.9	67.0	75.2	79.3	77.2	69.6	54.2	36.8	26.9
	5%	WB	14.0	15.2	28.1	39.5	53.3	64.2	69.0	69.1	61.4	46.8	31.9	20.1
		MCDB	14.9	16.7	31.2	46.3	62.1	72.6	76.4	74.5	66.5	50.6	33.7	20.9
	10%	WB	7.9	11.0	25.6	36.9	50.8	62.0	67.1	67.0	59.1	44.1	28.9	11.6
		MCDB	8.8	12.5	28.5	42.4	59.1	70.0	74.1	72.5	63.6	47.7	31.0	12.6
Mean Daily Temperature Range	5% DB	MDBR	14.2	17.1	18.8	16.1	18.2	18.3	15.8	14.9	15.9	13.7	13.2	13.9
		MCDBR	14.1	16.9	19.1	22.6	28.0	25.0	21.9	20.2	20.8	19.3	12.3	14.8
		MCWBR	13.4	15.5	15.2	13.8	15.1	13.6	11.8	11.5	12.7	12.7	9.9	14.6
	5% WB	MCDBR	13.4	16.2	17.7	21.6	24.6	22.3	19.1	17.8	18.8	17.8	11.1	14.9
		MCWBR	13.2	14.9	14.2	13.7	14.4	13.2	11.6	11.0	12.4	12.6	9.7	14.7
Clear-Sky Solar Irradiance	taub	0.280	0.310	0.381	0.489	0.551	0.521	0.520	0.448	0.385	0.390	0.336	0.288	
	taud	2.052	2.004	1.849	1.715	1.768	1.942	1.995	2.224	2.339	2.125	2.048	2.013	
	Ebn at Noon	216	245	246	233	226	236	234	246	250	213	187	184	
	Edh at Noon	24	35	51	66	66	56	52	40	31	31	25	22	
All-Sky Solar Radiation	RadAvg	388	757	1201	1469	1534	1707	1530	1290	1033	628	370	281	
	RadStd	33	41	81	122	165	185	191	124	90	56	43	34	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	+0.81	N/A	N/A	N/A	N/A	N/A	N/A	-248	+110	N/A

Nomenclature: See separate page